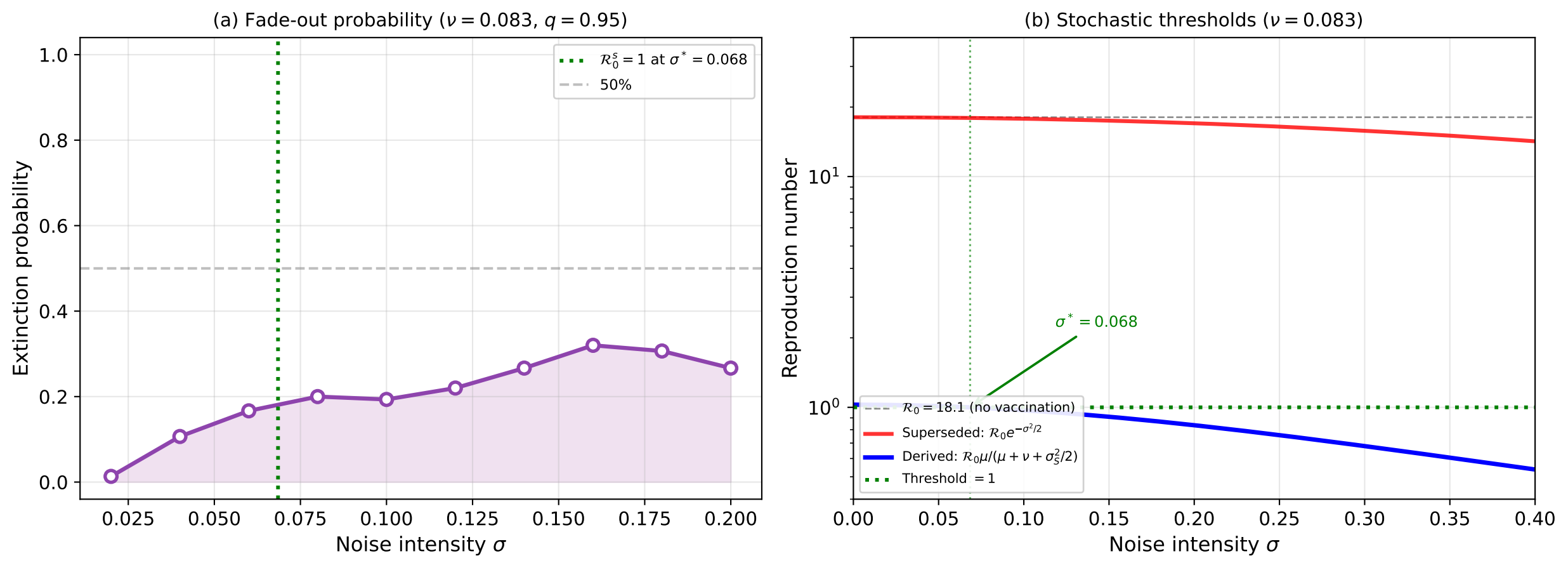
(a) Fade-out probability ($\nu = 0.083$, $q = 0.95$)
Extinction probability
Noise intensity σ
$\mathcal{R}_0^S = 1$ at $\sigma^* = 0.068$
50%
(b) Stochastic thresholds ($\nu = 0.083$)
Reproduction number
$\sigma^* = 0.068$
$\mathcal{R}_0 = 18.1$ (no vaccination)
Superseded: $\mathcal{R}_0 e^{-\sigma^2/2}$
Derived: $\mathcal{R}_0 \mu/(\mu + \nu + \sigma_S^2/2)$
Threshold $= 1$
Noise intensity σ